Weichen Huang

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Education

Xiamen University Expected June 2027
B.S. in Mathematics and Applied Mathematics (September 2023–present).
Current Rank: 1/39 GPA: 3.75/4.00.
Weighted Average: 89.61/100 Arithmetic Average: 90.42/100.
Junior-Year GPA: 3.98/4.00.
Junior-Year Averages: 94.80/100 (weighted); 95.19/100 (arithmetic).

Academic Interests

Homogeneous Dynamics; Teichmüller Theory and Dynamics; Ergodic Theory; Lie Groups; Riemann Surfaces; Geometry and Topology.

Selected Advanced Coursework

Analysis and Dynamics

- Functional Analysis (98)
  Uniform boundedness; open mapping; closed graph; Hahn–Banach; reflexivity; weak and weak-* topologies. Weak convergence; Riesz representation in Hilbert spaces; Lax–Milgram; compact operators.

- Ergodic Theory (97; graduate level)
  Focus on differentiable dynamical systems: hyperbolic fixed points and sets; stable manifolds; Hartman–Grobman theorem; symbolic dynamics; Smale horseshoe; Anosov toral automorphisms; solenoid attractors.

- Real Analysis (91)
  Lebesgue measure and integration; Egorov's and Lusin's theorems; dominated convergence theorem; Fubini's theorem; L^p spaces; convolution and Fourier transform.

- Mathematics Seminar (97)

Geometry and Topology

- Introduction to Differential Manifolds (100)
  Smooth manifolds; tangent spaces and vector fields; tensors and differential forms; Stokes' theorem. Lie groups and Lie algebras; affine connections, parallel transport, torsion, and curvature.

- Differential Geometry (93)
  Local theory of curves and surfaces; Frenet frames; first and second fundamental forms; shape operator and curvatures; intrinsic geometry and geodesics; isoperimetric inequality; Gauss–Bonnet theorem.

- General Topology (91)
  Metric and topological spaces; continuous maps and homeomorphisms; topological constructions; fundamental groups; covering spaces.

Complex Analysis and Geometry

- Several Complex Variables (99)
  Hartogs extension theorem; Cartan rigidity theorem; domains and weak domains of holomorphy; holomorphically convex domains; Cartan–Thullen theorem; plurisubharmonic functions; pseudoconvex domains.

- Complex Manifolds (97; graduate level)
  Complex manifolds; basic sheaf theory and Čech cohomology; Hermitian and Kähler manifolds.

Algebra

- Semisimple Lie Algebras and Representation Theory (95)
  Adjoint representation; Jordan–Chevalley decomposition; Killing form; representations of $\mathfrak{sl}_2(F)$; root-space decomposition; introduction to the classification of complex simple Lie algebras.

- Abstract Algebra (95)
  Groups and Sylow theorems; rings, principal ideal domains, and polynomial rings; modules and Schur's lemma; algebraic and transcendental field extensions; splitting fields; finite fields of order p^n.

Currently Enrolled

- Commutative Algebra (*in progress*)
  Text: Atiyah–Macdonald, *Introduction to Commutative Algebra*.
- Differential Topology (*in progress*)
  Transversality; degree theory; Poincaré–Hopf theorem.
- Introduction to Dynamical Systems (*in progress*)
  Ergodicity, recurrence and mixing; ergodic theorems; invariant and conditional measures. factors and joinings; Furstenberg's proof of Szemerédi's theorem.

Academic Projects & Expository Work

Geometric Modelling and Analysis of Dynamic Occlusion September 2025
Contemporary Undergraduate Mathematical Contest in Modeling (*CUMCM*), *Problem A; team project.*
First Prize, Fujian Division (team award).

- Developed a three-dimensional geometric model of dynamic occlusion based on viewing-cone containment.
- Provided geometric proofs and mathematical derivations supporting the model's key properties.
- Wrote the technical report, presenting modelling assumptions, geometric reasoning, and the team's numerical results.

Weyl Equidistribution and Unique Ergodicity Spring 2026
Mathematics Seminar, Xiamen University; 15-page expository note.

- Topics: weak-* topology of probability measures, invariant measures, unique ergodicity, irrational rotations, and Weyl equidistribution.
- Presented part of the material in the seminar.

Several Complex Variables and the Fatou–Bieberbach Theorem Fall 2025
Expository note.

- Exposition of holomorphic automorphisms in several complex variables and a dynamical proof of the Fatou–Bieberbach theorem, using invariant sets, conjugacy, and convergent solutions of functional equations.

Lie Groups and Homogeneous Geometry Fall 2026–present
Independent Reading; Zuo-Qin Wang's lecture notes.

- Current focus: Lie group actions, homogeneous spaces, and discrete subgroups of $\mathrm{PSL}(2, \mathbb{R})$.

Deep Learning Ongoing
Independent study using Zhonglei Wang's course materials [*This is a link :*)].

Summer Schools & Academic Activities

Tianyuan Mathematical Center in Southeast China July 2025
Topics: Introductory algebraic number theory; computation of birational invariants of rational functions and ordinary differential equations.

Westlake University Summer School July 2026
Topics: Calderón–Zygmund decomposition; arithmetic geometry.

Utrecht Summer School August 2026
Topics: Weierstrass preparation theorem in several complex variables; generic properties of smooth maps and embeddings.

Selected Honors & Awards

National Scholarship 2024–2025
Huang Xilie Scholarship (one of only 18 undergraduate recipients university-wide in 2025) 2024–2025
Outstanding Merit Student (university level) 2024–2025, 2025–2026

Languages & Technical Skills

Languages. Chinese – Native; English – IELTS 6.5 (retaking in November 2026).
French – Preparing for B1; Italian – Basic.
Technical skills. LaTeX; Python; MATLAB.